

Supplementary Material: Parameter Sensitivity

DeclareCascade: Explainable Structural Labelling of Process Concept Drifts

Balaktsis, Mavroudpoulos, Gounaris, Kougka

September 15, 2026

DeclareCascade uses a single fixed configuration ($\vartheta = 0.999$, $\xi = 0.01$) across datasets without per-dataset tuning. Figure 1 modifies ξ , s , $|B|$, and the localisation policy independently, holding all others at the default (dashed lines).

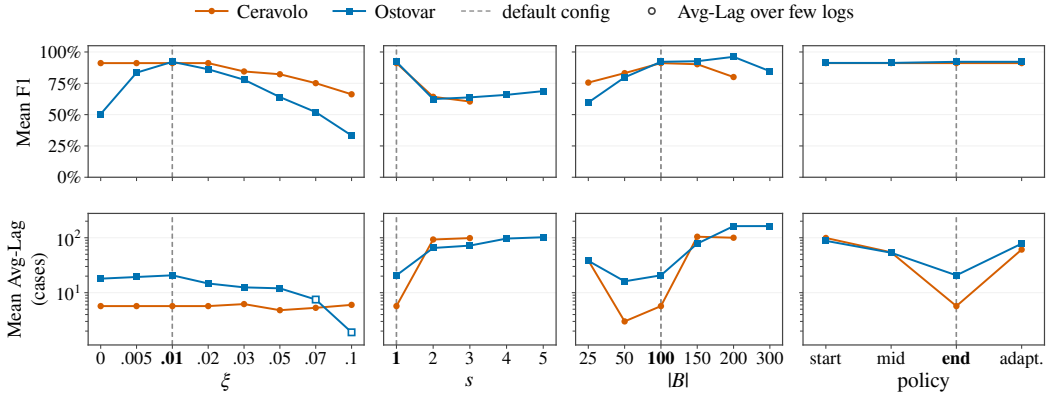


Figure 1: Parameter sensitivity sweep (each parameter varied independently; dashed line marks the default). *Top row*: macro-mean F1. *Bottom row*: mean Avg-Lag in cases (log scale); open markers indicate averages computed over fewer than half the logs.

Intensity threshold. $\xi = 0.01$ is the jointly optimal operating point. F1 is stable within $\xi \in [0.005, 0.02]$ on both datasets. Ostovar is more sensitive, with a sharp peak at 0.01 (0.505 at $\xi = 0$, rising to 0.922, then falling to 0.333 at $\xi = 0.10$), driven by its richer relational structure amplifying false positives at low thresholds and suppressing true detections at high ones; Ceravolo F1 is flat at 0.911 across $[0, 0.02]$ before gradually decaying. The apparent improvement in Ostovar lag at high ξ (1.9 at $\xi = 0.10$) is a coverage artefact: that average covers only 27 of 75 logs; the detector has stopped finding the harder drifts.

Trailing window. $s = 1$ is strictly best across both datasets. Increasing to $s = 2$ alone costs 0.27 F1 on Ceravolo and 0.30 on Ostovar, while inflating mean lag roughly $16\times$ ($5.7 \rightarrow 92.9$ cases) and $3\times$ ($20.7 \rightarrow 65.1$ cases) respectively. The mechanism is dual: a wider trailing window re-tests inside the same transition zone and re-fires on already-reported drifts, while simultaneously shrinking the number of testable boundaries. The sweep covers 10-batch and 30-batch streams; the separate $s = 2$ branch, activated only when ≤ 6 batches are observed, lies outside this regime.

Batch size. F1 peaks at $|B| = 100$ on Ceravolo (0.911) and $|B| = 200$ on Ostovar (0.961), making $|B| = 100$ the best joint compromise. Mean lag is minimised at $|B| = 50$ on both

datasets (3.0 and 16.1 cases), at the cost of moderate F1 reductions. Importantly, this trade-off does not extend indefinitely: reducing to $|B| = 25$ degrades *both* F1 and lag simultaneously (0.756/0.597 and 38.6/38.0 cases), because 25-case batches make per-batch proportions too noisy.

Localisation policy. F1 is essentially invariant to the choice of policy (exactly 0.911 on Ceravolo for all four; 0.913–0.922 on Ostovar), because the policy changes only which index within the firing batch is reported. Lag, however, spans a $17\times$ range (5.7 to 99.1 cases on Ceravolo). The *end* policy achieves the lowest lag on both benchmarks, a margin that reflects the benchmarks’ synthetic drift placement at exact batch boundaries: the last case of the firing batch coincides with the ground-truth change point by construction. The *adaptive* Manhattan-distance heuristic selects the wrong edge more often than not (lag 60.9 vs. 5.7 on Ceravolo), and does not outperform the fixed *end*.

Statistical test. F1 improves monotonically with ϑ from 0.99 (Ceravolo 0.762, Ostovar 0.702) to 0.999 (0.911, 0.922); at $\vartheta = 0.9999$ Ostovar gains marginally (0.939) but Ceravolo loses one TP, confirming $\vartheta = 0.999$. Using support alone achieves F1 0.907/0.935; confidence alone drops to 0.689/0.847; combining both improves Ceravolo by 0.004 but reduces Ostovar by 0.014 — complementary but support-dominated, with confidence retained for changes that redistribute event-level fulfilment without altering case-level support. The dual confirmation gate (intensity $\geq \xi \cdot |R|$ or label $\in \mathcal{L}$) recovers two additional TPs on Ostovar at a small precision cost (F1: 0.83→0.80); on Ceravolo the effect is negligible. The structural-label criterion thus acts as a safety net for low-intensity drifts, not a primary accuracy lever.